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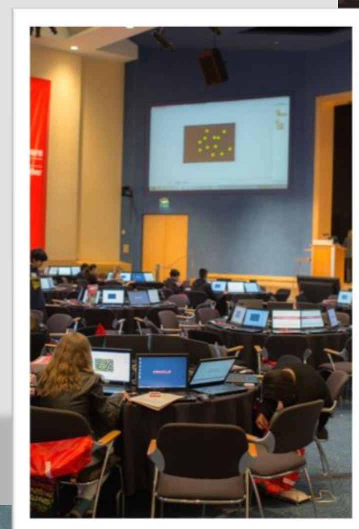
Database Foundations

4-1

Oracle SQL Developer Data Modeler



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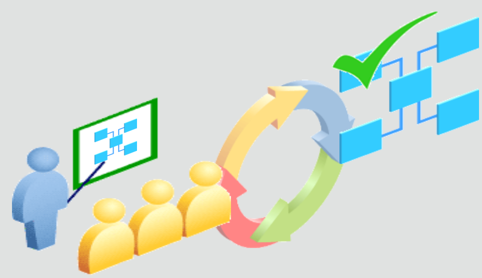
Objectives

- This lesson covers the following objectives:
  - Use Oracle SQL Developer Data Modeler to create:
    - Entities, attributes, and UIDs with correct optionality and cardinality
    - Supertype and subtype entities
    - Arc, hierarchical, barred, and recursive relationships

Introduction to Oracle SQL Developer Data Modeler

- Oracle SQL Developer Data Modeler offers a range of data and database modeling capabilities, enabling you to:
  - Capture business rules and information
  - Create process, logical, relational, and physical models
  - Store metadata information in XML files
  - Synchronize the relational model with the data dictionary



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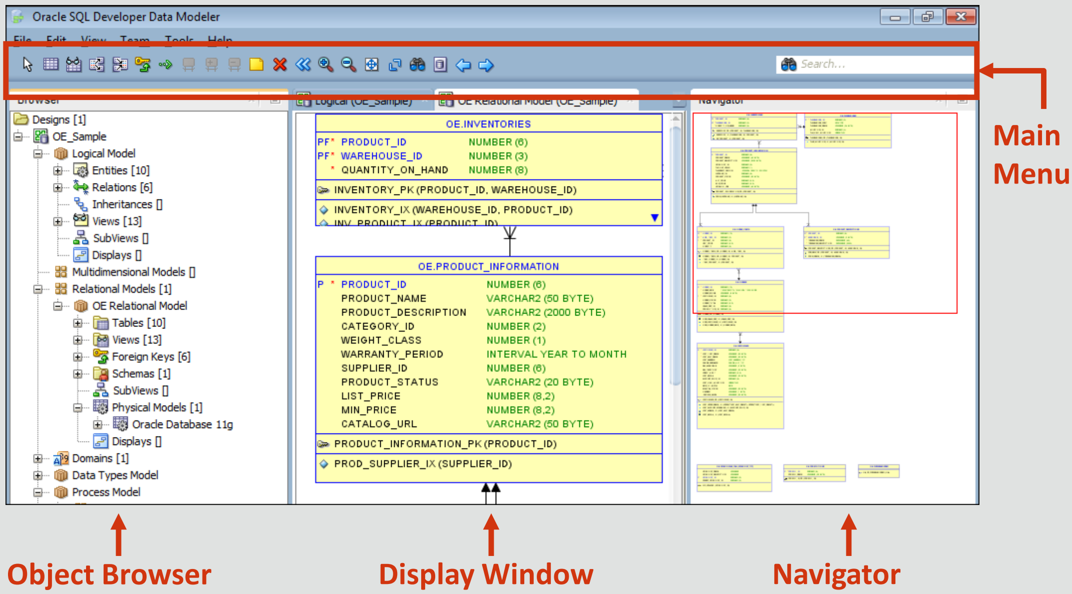
Oracle SQL Developer Data Modeler Interface: Example Overview



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Main Menu

The Main Menu contains some standard entries, in addition to entries for features that are specific to the Data Modeler. Icons under the menus perform actions relevant to the data model selected for display in the display window. For example, for a relational model, the icons include New Table, New View, Split Table, Merge Tables, New FK Relation, Generate DDL, Synchronize Model with Data Dictionary, and Synchronize Data Dictionary with Model.

Object Browser

The left side of the Data Modeler window has an Object Browser that displays data modeling objects in a hierarchical tree structure. You can select an object in the object browser by expanding the appropriate tree node or nodes and then clicking the object.

Navigator

It displays a graphical thumbnail representation of the view that is currently selected in the display window, to the right of the Object Browser.

Note: The screenshot in the slide displays a relational data model that has already been created.

Building an ERD by Using Oracle SQL Developer Data Modeler

a. Create entities

b. Add attributes and UIDs

c. Define relationship between entities

d. Set the source and target values for the relationship

Case Scenario: An Introduction

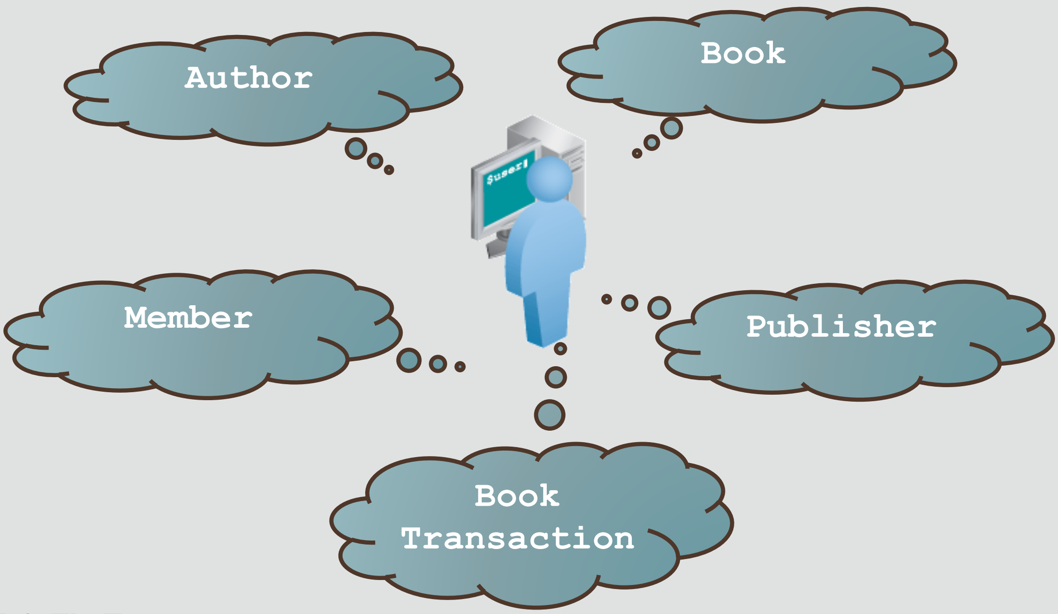
Faculty

Sean, I would like you to create a simplified library database to manage the number of reference books in our department. As a first step, can you build a logical model using Oracle SQL Developer Data Modeler that we have installed in our student machines?

Glad to. I'll start by identifying the entities and their attributes. After that, I can use the Oracle SQL Developer Data Modeler tool to build the logical model.

Student

Case Scenario: Identifying Entities



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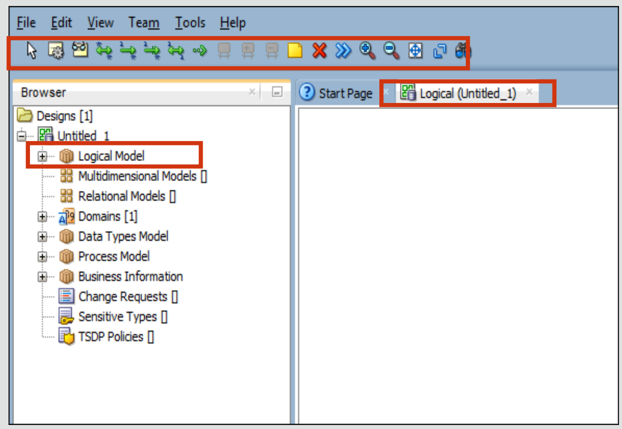
Building an Entity Relationship Diagram

1. Navigate to the Logical Tab
2. Create an Entity
3. Add Attributes to the Entity
4. Set the Unique Identifiers
5. Define the Relationship Between the Entities

Building an ERD: Step 1

- Navigate to the Logical tab
  - The first step in building an ERD in Oracle SQL Developer Data Modeler is to click the Logical tab. Notice that the toolbar changes to display tools specifically for working with ERDs
  - If you do not see the Logical tab, then perform the following steps:

1. Right-click the Logical Model in the browser
2. Select Show



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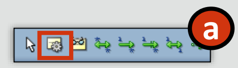
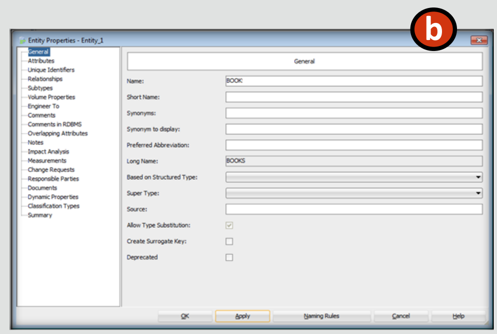
Building an ERD: Step 2

- Create an entity
  - To create an entity, perform the following steps:
    a. On the toolbar, click the New Entity tool, and then click anywhere in the white space of the Logical pane. The Entity Properties window appears
    b. In the Entity Properties window, enter the name of the entity. For the example in the slide, the entity name is "BOOK". Do not close the window after you have entered the entity name





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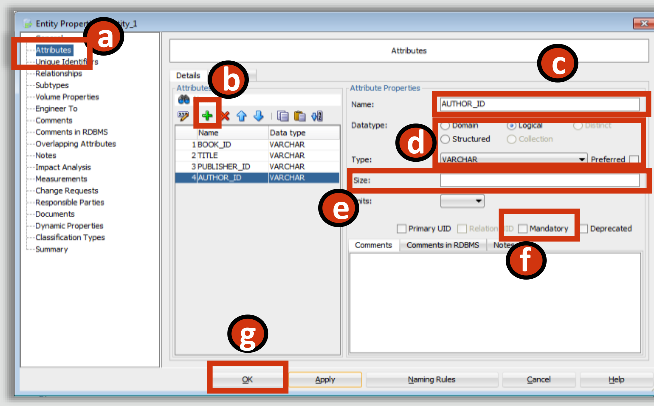
Building an ERD: Step 3

- Add attributes to the entity
  - To add attributes to the entity, perform the following steps:
    - a. Select Attributes in the navigator of the Entity Properties window
    - b. Click the Add an Attribute icon
    - c. In the Name field, enter a name for the attribute



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Data types will be covered in more detail later in the course. For dates, use the DATE type. For text use VARCHAR, and for numbers NUMERIC.

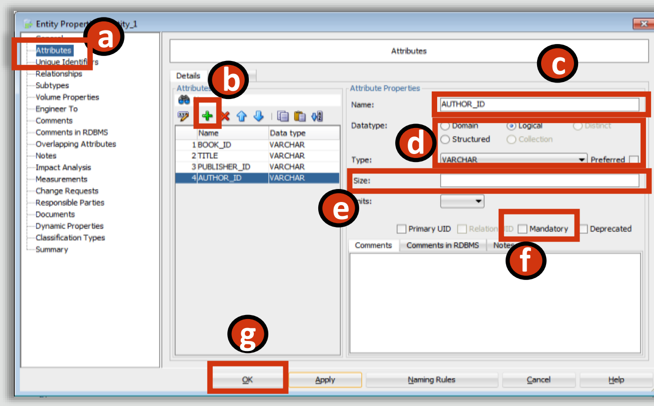
Attribute size is the maximum number of characters for VARCHAR types.

For NUMERIC datatypes: Precision is the maximum number of digits and Scale is the number of digits after the decimal point.

Logical types are not actual data types, but names that can be associated with native types.

Building an ERD: Step 3

- Add attributes to the entity
  - d. In the Data type field, select Logical, then required type. from the dropdown list
  - e. Enter the attribute size
  - f. If the attribute is mandatory, check the box
  - g. Click OK



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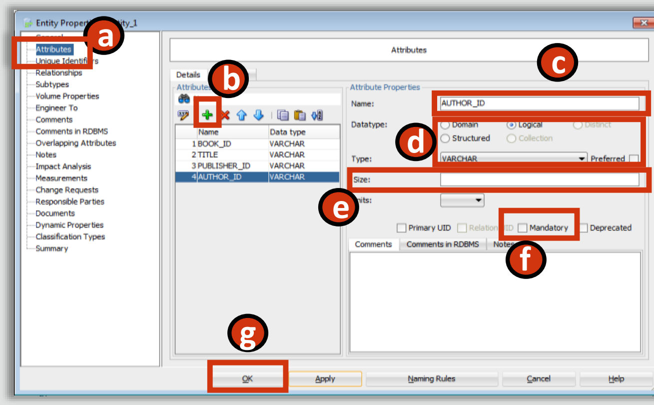
Building an ERD: Step 3

- Add attributes to the entity
  - Although Data types are not required in a logical model (they will not be shown on the ERD) adding them now will allow Data Modeler to convert them to actual SQL Data types when we engineer to the Physical (Relational) Model



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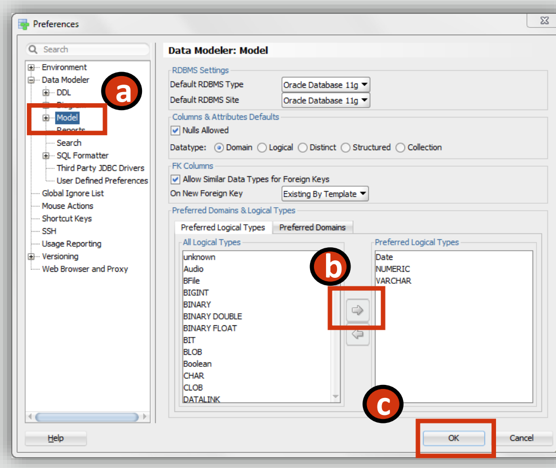
Logical types are not actual data types, but names that can be associated with native types.

Building an ERD: Step 3 cont.

- Set Preferred data types
  - You can set commonly used Logical data types as preferred by performing the following steps:
    a. Select Tools > Preferences > Data Modeler, and select the Model node
    b. Select the types from the All Logical Types section and move them to the preferred area by clicking the arrow
    c. Click OK



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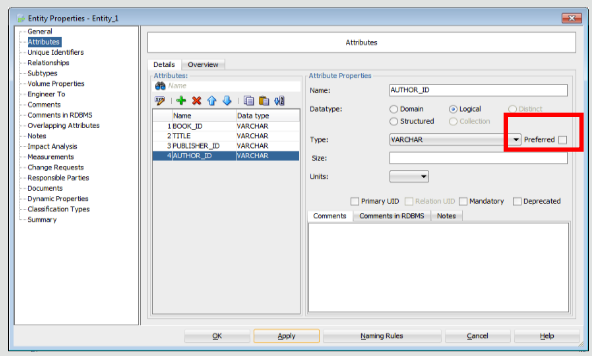
Building an ERD: Step 3 cont.

- Set Preferred data types
  - To view only the preferred data types, select the Preferred check box. This will limit the options displayed in the drop-down list for logical types



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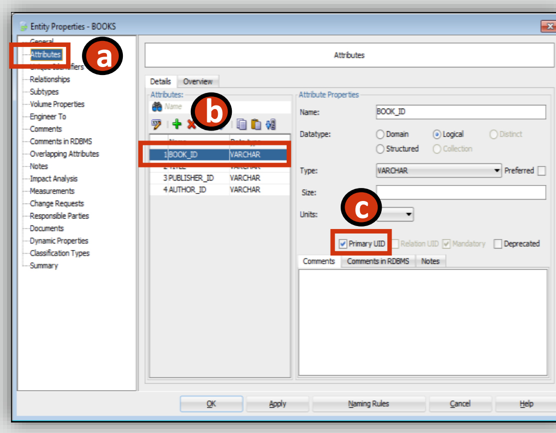
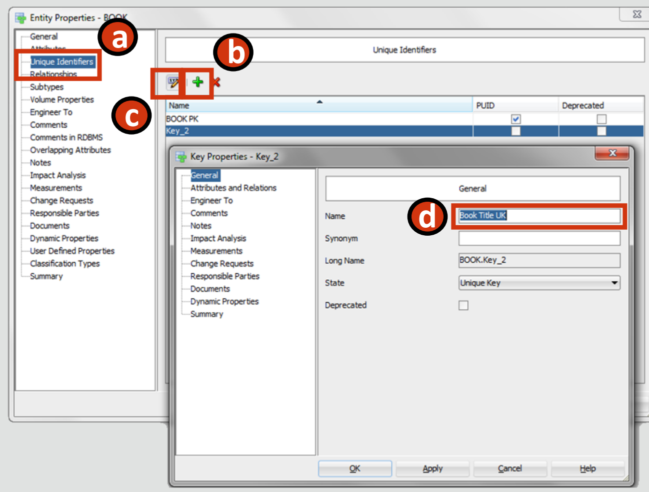
Building an ERD: Step 4

- Set Primary and Secondary UIDs
  - To set the Primary UID for the entity, perform the following steps:
    a. Select Attributes in the left navigator of the Entity Properties window
    b. Select the attribute that you want to assign as the Primary UID
    c. Select the Primary UID check box
- The attribute that you assign as primary UID is automatically also set to mandatory



Building an ERD: Step 4 cont.

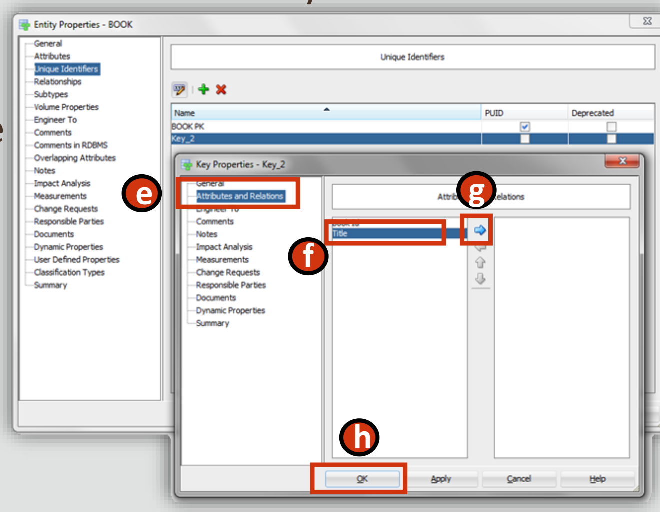
- Set Primary and Secondary UIDs
  - To set Secondary UIDs for the entity, perform the following steps:
    a. Select Unique Identifiers in the left navigator of the Entity Properties window
    b. Click the Add icon to add another UID
    c. Click the Properties icon
    d. Enter a name to identify the Secondary UID



(Continued on next slide)

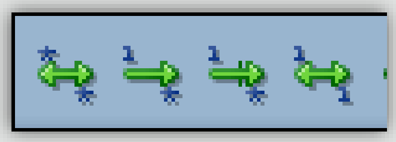
Building an ERD: Step 4 cont.

- Set Primary and Secondary UIDs
  - e. Select Attributes and Relations from the left navigator of the Key Properties window
  - f. Select the attribute to set as a Secondary UID
  - g. Click the arrow icon to move the attribute to the right-hand pane
  - h. Click OK



Building an ERD: Step 5

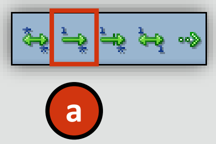
- Define the relationships between the entities

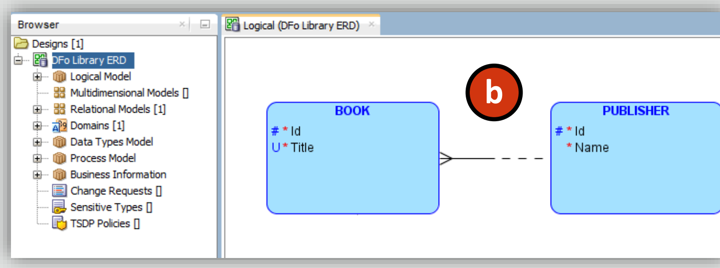


- The relationships available in Oracle SQL Developer are:
  - 1:1 (one-to-one)
  - 1:N (one-to-many)
  - 1:N Identifying Relationship (one-to-many barred relationship)
  - M:N (many-to-many)

Building an ERD: Step 5 cont.

- Define the relationships between the entities





- To define the relationships between entities in Oracle SQL Developer, perform the following steps:
  a. Click a relationship type on the toolbar
  b. Click the source entity and then click the target entity. The relationship is created

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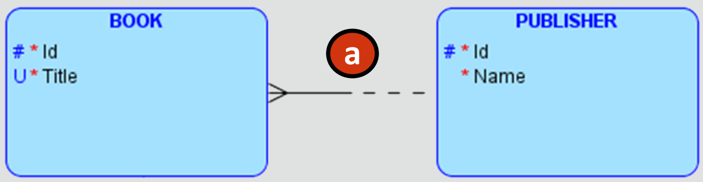
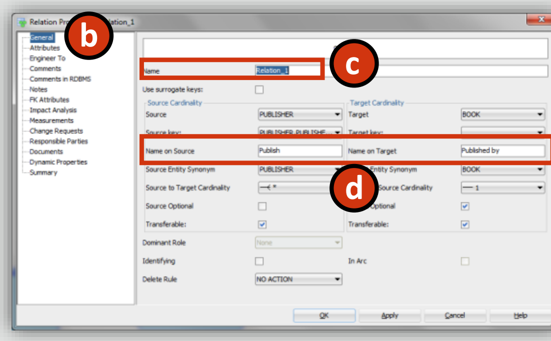
Building an ERD: Step 6

- Set the source and target values for the relationship



  – To set the source and target values for the relationship, perform the following steps:

a. Double-click the relationship in the diagram
b. Select the General property in the left navigator
c. Specify a name for the relationship
d. Specify the source and target names for the relationship

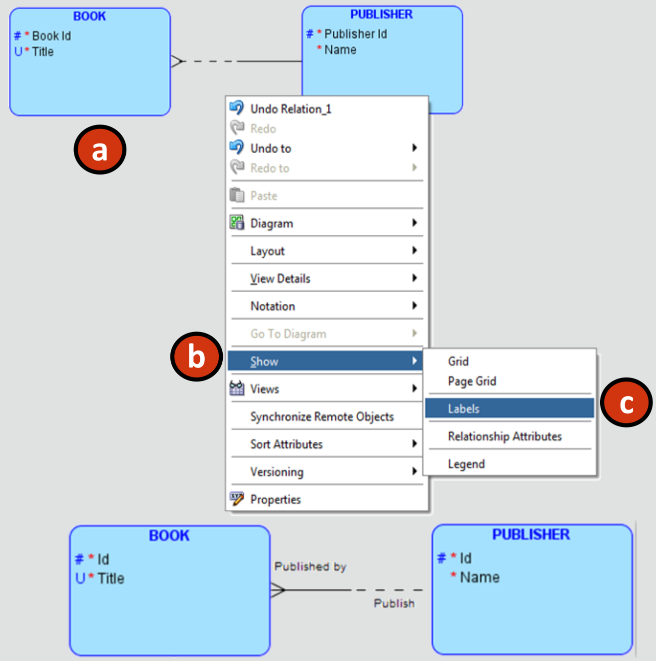


The source and target optionality and cardinality can be also be modified in the Relation Properties window.

To mark a relationship as non-transferable, uncheck the Transferable check box at either the source or the target.

Building an ERD: Step 6 cont.

- To display source and target relationship names:
  a. Right click on the whitespace of the diagram
  b. Select Show
  c. Click Labels



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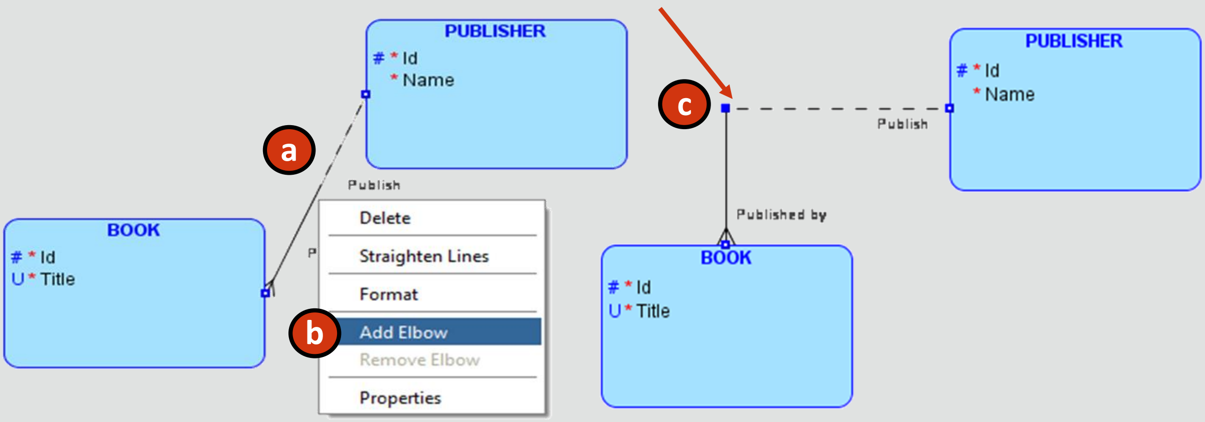
Building an ERD: Step 6 cont.

- To change the path of a relationship line:
  a. Right click on the relationship
  b. Select Add elbow
  c. Drag the center handle to the desired position



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Case Scenario: Entity Types

Faculty

Sean, I was wondering if we could include new types of membership categories such as:

- Student Membership
- Faculty Membership
- Corporate Membership

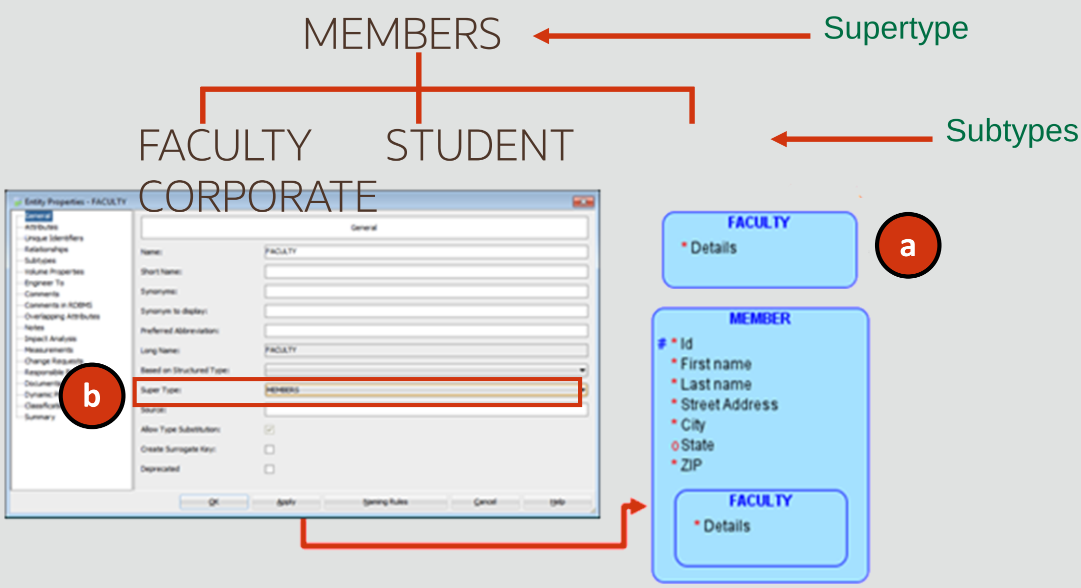
This can definitely be achieved. I can create a common entity that would hold membership details that are common to all the three membership categories. This would be a supertype entity. The specific membership categories would inherit the properties of the supertype entity, in addition to their own specific attributes. Hence, the specific membership category would be a subtype entity.

Student

Creating the Supertype Entity

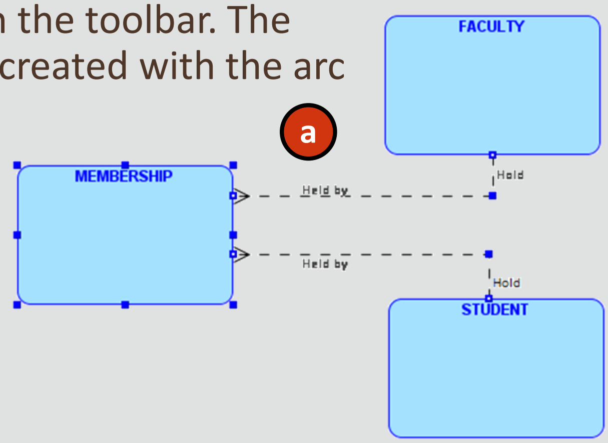
- To define an entity as a subtype in Oracle SQL Developer Data Modeler you need to ensure that the supertype exists. In this example, the super type is MEMBER. Perform the following steps to create the FACULTY subtype:
  a. Double-click the entity that you want to make a subtype. For the example in the slide, you want to make FACULTY a subtype of the MEMBER supertype. Double-click FACULTY
  b. Select the MEMBER supertype entity from the Super Type list, and click OK. The FACULTY entity is now a subtype of the MEMBER supertype and will inherit all the attributes of the supertype

Creating the Supertype Entity



Creating the Arc Relationship

- To create an exclusive relationship in Oracle SQL Developer Data Modeler, perform the following steps:
  a. Hold the ctrl key and select the intersecting entity and both relationships on which you want to create the Arc relationship
  b. Click the New Arc icon in the toolbar. The exclusive relationship is created with the arc



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The business rule for this relationship is read as follows: "Each MEMBERSHIP either may be held by one and only one STUDENT or may be held by one and only one FACULTY."

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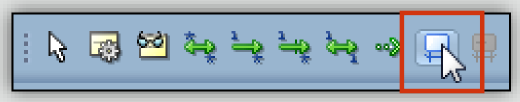

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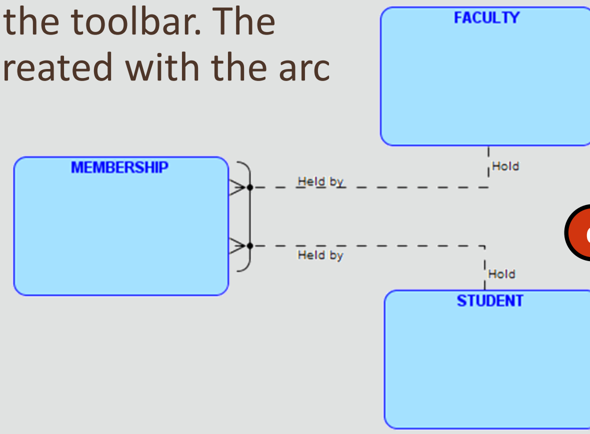

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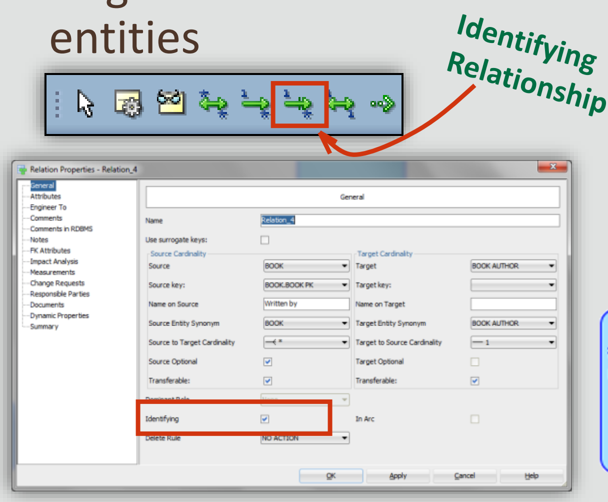
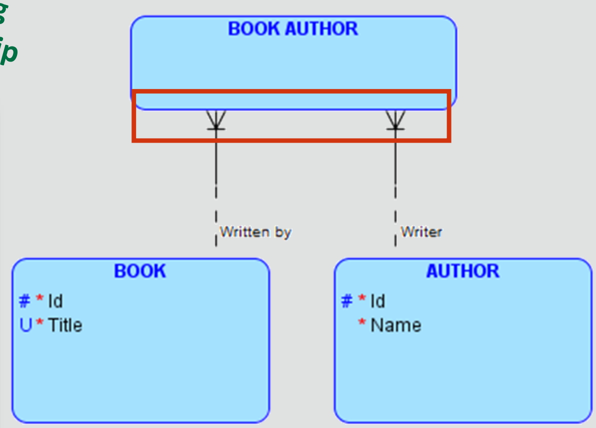
The business rule for this relationship is read as follows: "Each MEMBERSHIP either may be held by one and only one STUDENT or may be held by one and only one FACULTY."

Creating the Barred Relationship

- To add a barred relationship select Identifying Relationship from the toolbar, and click the source and target entities to add the relationship between the entities





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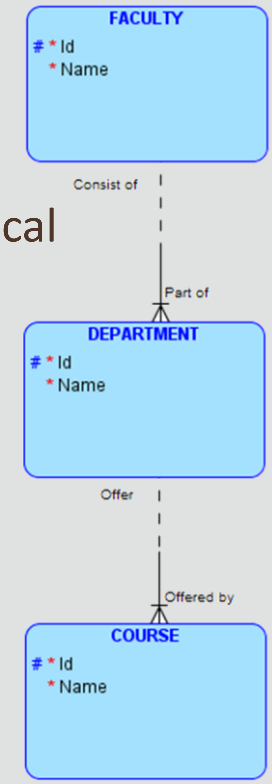
Creating the Hierarchical Relationship

- Hierarchical data can be represented as a set of 1:N relationships (or 1:N Identifying)
- In this example, a University has a hierarchical structure:
  - A FACULTY can consist of one or more DEPARTMENTS
  - A DEPARTMENT can offer one or more COURSES
- The UIDs for a set of hierarchical entities can be propagated through multiple relationships by making the relationships Identifying



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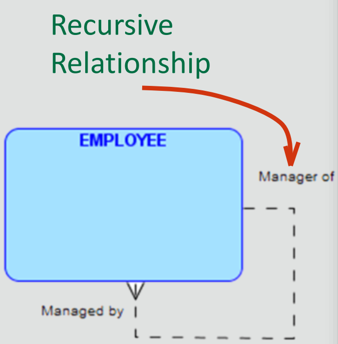
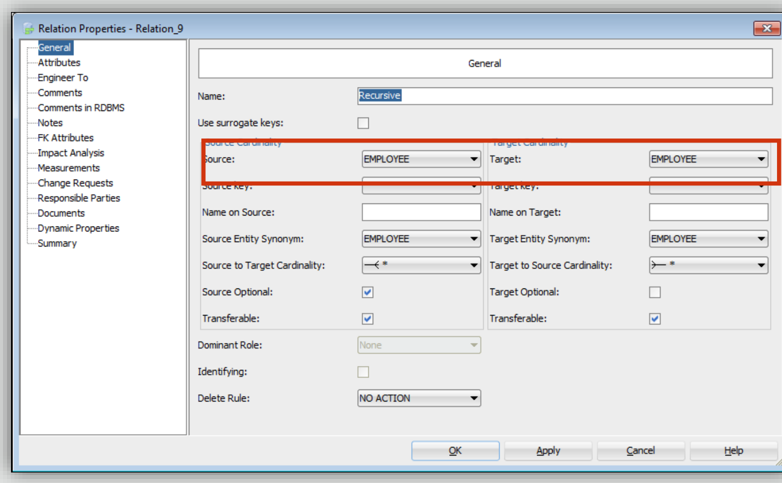
In this example, the UID of COURSE is a composite UID consisting of Course Id, Department Id and Faculty Id.

Creating the Recursive Relationship

- To add a Recursive Relationship, select the required relationship from the toolbar as normal, then click on the entity to make it the source, and click on the same entity a second time to make it the target





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The business rule for this relationship is read as follows: "An employee may be a manager of one or more employees“. "An employee may be managed by one and only one employee“

Project Exercise

- DFo_4_1_Project
  - Oracle Baseball League Store Database
  - Creating a Logical Data Model

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Summary

- In this lesson, you should have learned how to:
  - Use Oracle SQL Developer Data Modeler to create :
    - Entities, attributes, and UIDs with correct optionality and cardinality
    - Supertype and subtype entities
    - Arc, hierarchical, barred, and recursive relationships

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